Customer Churn Prediction System

End-to-End ML System with CRISP-DM & MLOps (MLflow)

Hector Alejandro Vargas Gutierrez

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Executive Summary

- **Problem:** Predict customer churn in a retail banking environment.
- **Objective:** Identify customers at risk of closing their accounts to support proactive retention strategies.
- **Best Model:** XGBoost optimized using Bayesian hyperparameter tuning (Optuna TPE).
- **Optimization Metric:** Average Precision (PR-AUC) = 0.698 (5-Fold Cross-Validation).
- **Test Performance:** PR-AUC = 0.734, ROC-AUC = 0.874, Accuracy = 87.3%.
- **Best Configuration:** RobustScaler, 1,403 trees, learning rate = 0.028, subsample = 0.97, column sample rate = 0.52, scale_pos_weight = 1.19.
- **Production System:** End-to-end ML pipeline with feature engineering, MLflow experiment tracking, (model deployment API, and monitoring framework - PENDING).

Business Problem

- Customer churn in banking leads to significant revenue loss and higher acquisition costs
- Losing customers reduces deposits, transaction volume, and product cross-sell opportunities
- Early detection enables targeted retention strategies (offers, engagement, support)
- Objective: Predict probability of customer churn (*Exited*)

CRISP-DM Methodology

Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment

Dataset Overview

Dataset Summary

- Customers: 10,000
- Features: 17 input features + 1 target variable (Exited)
- Target: Churn (0 = stay, 1 = leave)
- Data Types: Numerical, categorical, and binary features
- Missing Values: None
- Problem Type: Binary Classification

Objective

Predict customer churn and identify the main drivers influencing customer departure.

Data Characteristics

Numerical Features

- CreditScore
- Age
- Tenure
- Balance
- EstimatedSalary
- Point Earned

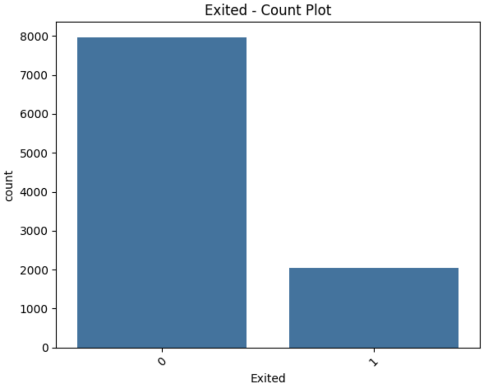
Categorical / Binary Features

- Geography
- Gender
- Card Type
- HasCrCard
- IsActiveMember
- Complain

Target Variable

`Exited`: 0 = Retained Customer, 1 = Churned Customer

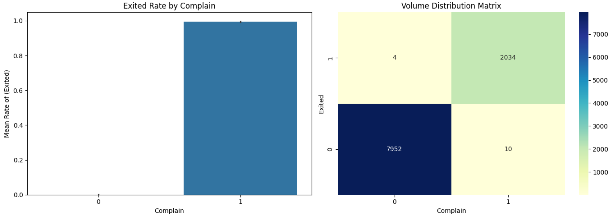
Target Distribution



Observation

- Approximately 20% of customers churned
- Dataset is moderately imbalanced
- Evaluation focused on PR-AUC
- Class weighting applied in XGBoost

EDA: Customer Complaints and Churn

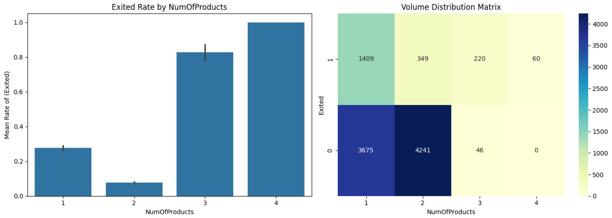


Key Finding

Customers who submitted complaints are substantially more likely to churn.

- Complaint status shows one of the strongest associations with churn.
- Customer service interactions appear to be a critical retention factor.
- The distribution suggests a strong positive correlation

EDA: Number of Products and Churn

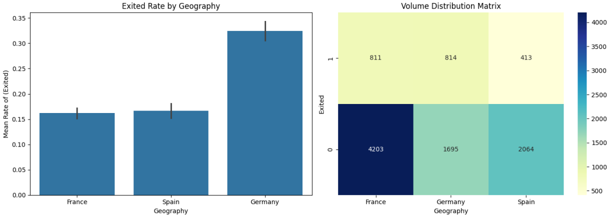


Key Finding

Customers holding 3 or 4 products exhibit noticeably higher churn rates.

- Customers with 1–2 products are generally more stable.
- High-product customers may require targeted retention strategies.

EDA: Geography and Churn

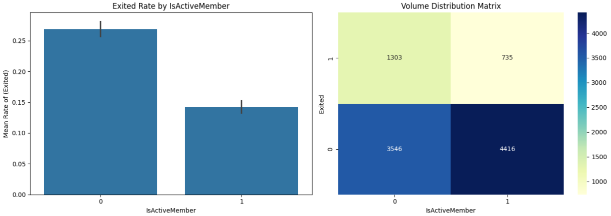


Key Finding

Customers in Germany exhibit significantly higher churn rates compared to France and Spain.

- Geographic location is a strong predictor of customer retention.
- Country-specific retention initiatives may be warranted.

EDA: Active Membership and Churn



Key Finding

Inactive customers churn at much higher rates than active members.

- Customer engagement strongly correlates with retention.
- Strategies to increase activity could reduce churn effectively.

Data Preparation

- No missing values detected; no imputation required.
- One-Hot Encoding applied to categorical features with `drop='first'` to avoid multicollinearity: `Card Type`, `NumOfProducts`, `Geography`, `Gender`.
- Standardization applied to continuous features: `Balance`, `Point Earned`, `CreditScore`, `Age`, `Tenure`, `Satisfaction Score`, `EstimatedSalary`.
- Binary features retained as-is: `HasCrCard`, `IsActiveMember`.
- Binary feature `Complain` was excluded from the analysis due to its strong correlation with the target variable, which could introduce target leakage and artificially inflate model performance.
- Train, validation, and test split performed after preprocessing.

Baseline Model Comparison

- Several classification algorithms were evaluated using default hyperparameters:
  - Logistic Regression
  - Decision Tree
  - Random Forest
  - XGBoost
- Model performance was primarily assessed using Precision–Recall AUC (PR-AUC).
- PR-AUC was preferred over Accuracy because customer churn prediction is an imbalanced classification problem.
- In churn management, correctly identifying churners is more important than correctly classifying the majority non-churning customers.
- Random Forest achieved the highest baseline PR-AUC (0.733), followed closely by XGBoost (0.692).

Model Selection for Optimization

- Although Random Forest achieved the best performance with default settings, XGBoost was selected for further development.
- The difference in baseline PR-AUC was relatively small (0.733 vs. 0.692).
- XGBoost provides:
  - More extensive hyperparameter tuning capabilities.
  - Gradient boosting optimization that often outperforms bagging methods after tuning.
  - Built-in regularization to improve generalization.
  - Better handling of complex feature interactions.
- Therefore, XGBoost was considered the model with the highest potential for performance improvement during the modeling phase.

Feature Importance Analysis

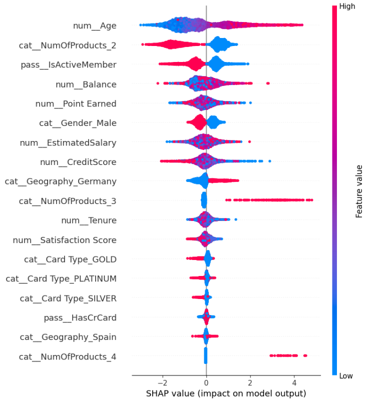
- Feature importance was evaluated using the selected XGBoost model.
- Customer product ownership was the strongest predictor of churn:
  - `NumOfProducts_2` (25.8%)
  - `NumOfProducts_3` (15.4%)
  - `NumOfProducts_4` (8.4%)
- Customer engagement also showed high predictive power:
  - `IsActiveMember` (8.7%)
  - `Age` (6.6%)
- Geographic location contributed significantly:
  - `Geography_Germany` (7.5%)
  - `Geography_Spain` (2.7%)
- The five most important features represent approximately 66% of the total feature importance assigned by the model.
- Card type variables and credit card ownership contributed relatively little to churn prediction.

Key Business Insights

- Product ownership is the primary driver of customer churn.
- Active customers are substantially less likely to leave the bank.
- Churn behavior varies across geographic regions, particularly in Germany.
- Older customers exhibit different retention patterns than younger customers.
- Credit card ownership and premium card tiers have limited impact on churn behavior.
- Retention initiatives should prioritize:
  1. Customer engagement programs.
  2. Product portfolio optimization.
  3. Region-specific retention strategies.

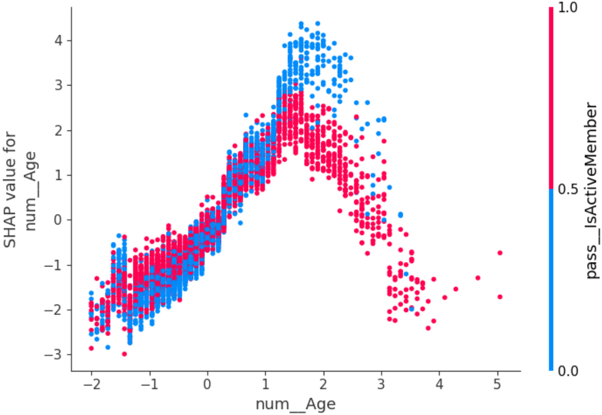
SHAP Model Interpretation

- SHAP values were used to explain feature contributions to churn predictions.
- Factors increasing churn probability:
  - Higher customer age
  - Customers with 3 or 4 products
  - German customers
  - Higher account balances
- Factors reducing churn probability:
  - Active membership
  - Customers with exactly 2 products
  - Higher credit scores



SHAP summary plot for the final XGBoost model.

Age Effect on Churn Prediction: SHAP Dependence



SHAP dependence plot for `Age`, colored by `IsActiveMember`.

Age Effect on Churn Prediction: Key Findings

- Customer age exhibits a strong non-linear relationship with churn predictions.
- A transition occurs around standardized age ≈ 2.0, where the SHAP contribution changes from positive to negative.
- Younger customers tend to significantly decrease the predicted probability of churn.
- Middle-age customers tend to increase the predicted probability of churn.
- Older customers tend to decrease the predicted probability of churn.
- Membership status moderates this effect:
  - Inactive customers amplify the churn risk.
  - Active customers further reduce the churn risk.

Business Insight

Prioritize retention campaigns for middle-age customers, while older and younger customers represent a relatively low-risk segment.

Sequential Feature Selection Results

Methodology

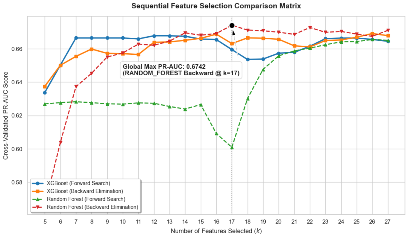
Sequential Feature Selection (Forward and Backward) was applied to Random Forest and XGBoost models, optimizing PR-AUC through 5-Fold Cross-Validation.

Best Configurations

Model	Feat.	PR-AUC
RF + Backward	**17**	**0.6742**
XGB + Backward	16	0.6688
XGB + Forward	12	0.6678
RF + Forward	27	0.6656

Observations

- Best overall result achieved by RF Backward SFS.
- Peak performance reached with 17 selected features.
- Backward elimination consistently outperformed forward selection.
- Engineered interactions remained valuable even as feature count increased.



Best Feature Subset Analysis

Winning Model

Random Forest with Backward Sequential Feature Selection achieved the highest PR-AUC (**0.6742**) using a subset of **17 predictors**.

Selected Feature Groups

Category	**Selected Features**
Demographics	Gender, Germany, Spain
Products	NumOfProducts_1, 2, 3, 4
Customer Value	Estimated Salary, Point Earned, Satisfaction Score
Interactions	Age×IsActive, CreditScore×Age, Inactive×Balance
Ratios	Balance per Product, CreditScore per Age, Products per Tenure
Card Type	Silver Card

Key Findings

- Product ownership variables appeared in every high-performing subset.
- Interaction features contributed more than raw customer attributes.
- Financial ratios captured customer behavior better than standalone balance measures.
- Geography remained a persistent source of churn heterogeneity.
- Satisfaction and loyalty indicators added complementary predictive power.

Modeling Strategy and Hyperparameter Optimization

Optimization Framework

Optuna's Tree-structured Parzen Estimator (TPE) was used to maximize **Average Precision (PR-AUC)** through a Bayesian search of **5,000 trials**.

- End-to-end pipeline including:
  - Dynamic feature engineering
  - Scaling strategy selection (Standard, MinMax, Robust)
  - Encoding strategy selection
  - Model selection (Random Forest vs XGBoost)
- Evaluation performed using **5-fold Stratified Cross Validation**.
- Optuna's **MedianPruner** stopped poorly performing trials early, reducing computational cost.
- Optimization target: **PR-AUC**, chosen because churn prediction is an imbalanced classification problem.

Experiment Tracking with MLflow

Reproducible Optimization Workflow

MLflow was integrated with Optuna to automatically track all hyperparameter optimization experiments.

- SQLite backend used for persistent experiment storage.
- Parent-child run structure:
  - Parent run stores final model results.
  - Child runs store individual Optuna trials.
- Automatic logging of:
  - Hyperparameters
  - Cross-validation scores
  - Test-set metrics
  - Feature schema
  - Serialized production pipeline
- Ensures full reproducibility and auditability of model development.

Best Model Performance

Metric	Value
Cross-Validation PR-AUC	0.698
Train PR-AUC	0.797
Test PR-AUC	0.734
Accuracy	0.873
Precision	0.749
Recall	0.564
F1 Score	0.643
ROC-AUC	0.874

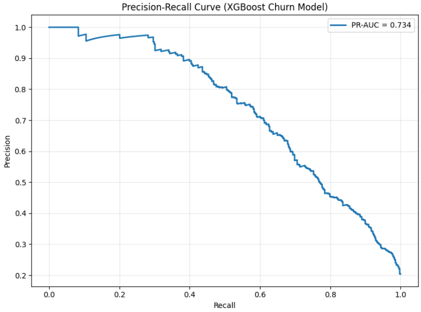
Model Generalization

The final XGBoost model achieved a Test PR-AUC of 0.734, with a limited gap compared to training performance (Train PR-AUC = 0.797), indicating good generalization on unseen customer data.

Model Diagnostics: Precision-Recall Curve

Champion Model Performance

- **PR-AUC Metric: 0.734**
- **Direct Alignment:** Evaluates the exact metric targeted during the 5000-trial Bayesian search space loop.



Why PR-AUC Over ROC-AUC?

Because our churn target is highly imbalanced, ROC-AUC can present an overly optimistic view by factoring in true negatives. The Precision-Recall curve focuses strictly on the minority class, tracking false positives and false negatives without distortion.

Business Trajectory Context

Allows executive stakeholders to dynamically tune model probability thresholds to balance marketing outreach costs (Precision) vs. capturing actual churning clients (Recall).

Hyperparameter Search Insights

Optimal Search Outcome

Optuna's Bayesian optimization identified XGBoost as the most effective architecture after 5,000 trials, maximizing Average Precision (PR-AUC).

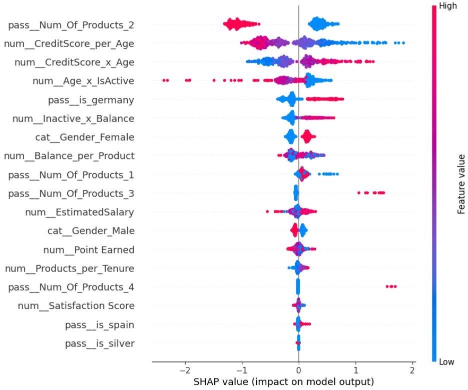
- **Model Selection:** XGBoost consistently achieved the highest PR-AUC compared with Random Forest alternatives.
- **Preprocessing:** Robust scaling provided the most stable performance under feature distribution variability.
- **Class Imbalance:** A moderate positive-class weight (`scale_pos_weight = 1.19`) improved minority-class detection.
- **Optimization Behavior:** The search converged around:
  - learning rate ≈ 0.028
  - subsampling ≈ 0.97
  - feature sampling ≈ 0.52

Key Drivers of Churn Prediction

Global SHAP Interpretation

SHAP values quantify the contribution of each feature to the model output across all customers.

- **Product Relationship** Number of products is the strongest predictor of churn risk.
- **Financial Profile** Credit score interactions with age provide strong predictive signals.
- **Customer Engagement** Inactivity combined with balance identifies disengaged profiles.
- **Lower Impact Variables** Salary, satisfaction score, and card type have limited influence.



Individual Prediction Explanation Using SHAP

SHAP Waterfall Interpretation

SHAP values explain how individual features move the prediction from the average model output toward the final customer risk score.

Example Customer Explanation

- Features increasing churn probability:
  - CreditScore × Age interaction
  - CreditScore per Age
  - Inactive balance interaction
  - Age and activity relationship
- Features reducing churn probability:
  - Balance per product
  - Point earned
  - Gender contribution

Business Value:
Provides transparent explanations that can support targeted customer retention strategies.